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Bayesian neural network prediction of lens and source parameters for strong lensing samples from Hyper Suprime-Cam Survey

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ABSTRACT

Strong gravitational lenses probe both the matter distribution around lens galaxies as well the properties of the source galaxies that get lensed. Traditional modelling approaches to sample from the posterior distribution of parameters given multi-wavelength imaging data are computationally expensive. We present a Bayesian neural network (BNN), trained on realistic simulations of strongly lensed systems, to rapidly estimate key lens and source parameters as well as the errors on the Einstein radius, mass ellipticity components, source position, and its effective radius. We extend this framework to also output the full posterior distributions of parameters that concern the lens mass distribution. Our work represents one of the first efforts in strong lens modelling to recover the entire covariance matrix of model parameters using a Bayesian Neural Network and can offer a scalable solution for analysing the large volume of lensing data expected from future surveys such as the Rubin Legacy Survey of Space and Time.

Key words: gravitational lensing: strong – methods: data analysis

1 INTRODUCTION

A background galaxy can get multiply imaged and anisotropically magnified due to the presence of mass distribution around a foreground galaxy, upon sufficient alignment. This strong gravitational lensing phenomenon is a powerful astrophysical probe for studying the mass distribution around galaxies as well as the properties of the magnified source galaxies which may be otherwise faint to detect. If a transient is found in the background source, the time delays between the images can further constrain cosmological parameters, such as the expansion history of the Universe.

Modelling of strong lenses have previously involved just the positions and fluxes of source images identified in the system. However, the use of multi-wavelength images of strongly lensed systems can be more constraining. The colors of lens and source galaxies are typically different and any correlations between lens light and mass can be used to provide informed priors on the mass distribution in the gravitational lens.

The traditional modelling of strong lens systems with such multi-wavelength images (e.g., Nightingale et al. 2021; Birrer & Amara 2018; Suyu & Halkola 2010) involves the determination of the posterior distribution of parameters given the data, often using techniques such as Markov Chain Monte Carlo (MCMC) sampling. However, such methods rely on the multiple evaluations of the likelihood which compares multiwavelength model images based on the parameters to the imaging data. These evaluations are therefore computationally expensive. In addition the likelihood surface can be quite complex in the multi-dimensional model parameter space and thus require substantial manual intervention to ensure fast convergence.

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The evaluation of the posterior distribution of parameters presents a major bottleneck for performing lens modelling and scaling it to the large volume of data expected from upcoming wide-field surveys such as Euclid (e.g., Euclid Collaboration et al. 2025), LSST (e.g., Ivezić et al. 2019) and Roman (e.g., Eifler et al. 2021). To address this limitation, recent efforts have focused increasingly on developing automated and efficient alternatives, particularly those that involve machine learning to process large data sets rapidly but reliably (e.g., Hezaveh et al. 2017; Pearson et al. 2019, 2021; Schuldt et al. 2021, 2023a,b; Gentile et al. 2023b). Analysing a large number of lens systems simultaneously is essential for enabling population-level studies. This statistical power is especially crucial given the expected scale of upcoming surveys, which will detect thousands of strong lensing systems.

Gravitational lensing introduces very specific features in the imaging data. Lens systems with larger Einstein radii result in iamges which are further separated, the presence of an asymmetry in the lens system can change the locations and configuration of the lensed images formed in the system. Changing the locations of the source can change the multiplicity as well as the shapes of the observed lensed images. If such features can be gleaned from the imaging data, the model parameters can be inferred.

Convolutional Neural Networks (CNNs) have emerged as powerful feature extraction tools from images and therefore have been employed for fast and automated inference in gravitational lens modelling. Hezaveh et al. (2017) demonstrated that Convolutional Neural Networks (CNNs) could be used to estimate lens model parameters, such as the Einstein radius, from strong lensing images. Perreault Levasseur et al. (2017) extended this approach by applying a Bayesian framework to quantify uncertainties in these inferred parameters. Both of these studies primarily relied on high-resolution

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imaging data, similar to what is obtained from the Hubble Space Telescope.

To adapt machine learning methods to the upcoming surveys, Pearson et al. (2019) explored CNN-based inference using simulated lens images adapted to the expected quality of ground-based LSST data and the space-based Euclid mission. Schuldt et al. (2021) proposed a neural network for parameter estimation using simulations matched to HSC-like image quality. Later, Schuldt et al. (2023a) developed a more advanced residual network (ResNet) for the same purpose, which was subsequently applied to real SuGOHI lenses in Schuldt et al. (2023b).

In Gawade et al. (2025, henceforth, referred to as G25), we implemented a convolutional neural network (CNN) to estimate lens mass parameters from optical imaging data. The model was trained and tested on both simulated datasets and real strong lensing systems to evaluate its performance. We constructed training datasets with and without lens light to assess the impact of light contamination. We also studied the possible correlation of the predicted lens mass parameters with the lens light parameters in the presence of lens light. To compare model outputs with traditional methods, we applied both our CNN and the MCMC-based YL pipeline to a set of 182 SuGOHI lenses. For benchmarking, we also included a subset of 10 lenses previously modelled in independent studies to evaluate agreement in parameter recovery. The aim was to analyse how modelling assumptions, data complexity and lens-light subtraction influence parameter estimation, particularly in challenging ground-based observations.

Once trained on large, representative datasets, CNNs can rapidly predict key lensing parameters, such as the Einstein radius and mass ellipticity directly from imaging data, offering significant speed advantages over traditional methods. However, standard CNNs provide only point estimates and do not quantify the errors associated with these predictions. Such uncertainty is often assessed based on the comparison of network predictions on simulated test samples. However, errors quantified in such a manner may not easily generalize to errors in the network predictions when applied to real data (see G25).

Bayesian Neural Networks (BNNs) offer a compelling extension to standard CNNs for uncertainty quantification. BNNs can be used to infer the approximations of posterior distributions of parameters rather than single-point predictions. This makes BNNs particularly suited for scientific applications where uncertainty estimation is crucial.

Recent work has demonstrated the application of BNNs to automated lens modelling pipelines. For instance, the LEns MOdelling with Neural networks (LeMoN) framework (Gentile et al. 2023a) leverages a convolutional Bayesian neural architecture to infer the parameters of the Singular Isothermal Ellipsoid (SIE) model given images of a gravitational lens system.

In addition to forward modeling, BNNs have also been integrated into hierarchical inference frameworks to enable statistically consistent population-level parameter estimation. Wagner-Carena et al. (2021) employed BNNs in a simulation-based inference setting to estimate hyperparameters of lens populations, addressing the mismatch between training and test data distributions. This is accomplished by marginalizing the posteriors of individual lenses inferred by the BNN, thus accounting for heterogeneity and measurement uncertainty in a principled Bayesian fashion.

Furthermore, Park et al. (2020) explored the use of BNNs for time-delay cosmography, modelling hundreds of lens systems to estimate the Hubble constant. Their approach demonstrated that, by incorporating predictions along with uncertainties from BNNs, one can preserve the accuracy of standard lens modelling pipelines while gaining substantial computational speed, which is an essential feature for interpreting large samples from future surveys. Overall, the use of BNNs in strong lens modelling offers a path toward scalable and automated inference pipelines, positioning them as indispensable tools for next-generation lensing studies.

In this work, we extend the approach presented in our previous work G25 to incorporate additional parameters relevant to the modelling of strongly lensed systems as well as quantify the uncertainty in the determination of these parameters using a BNN. The new parameters include properties of the source such as the position of the unlensed source compared to the lens galaxy as well as its effective radius. These parameters enhance the model's ability to capture more complex characteristics of the gravitational lens systems. Compared to previous studies, we also present an extended BNN that can recover the full covariance matrix of model parameters, providing functionality similar to the traditional methods but in a much more rapid fashion. Such a neural network can infer parameters that characterize the posterior distribution of the model parameters given the imaging data. We expect that such BNNs will become crucial to process the large number of images of gravitationally lensed systems which will be detected from upcoming ground-based surveys such as Rubin LSST.

2 CONSTRUCTION OF TRAINING DATA

Machine learning models have a large number of parameters which have to be adapted based on the intended outputs given an input. The values of the parameters that get finally adapted can be strongly influenced by the realism and the diversity of the data it is trained on. In this section, we briefly present the procedure adopted to generate the synthetic training and validation datasets used for training our network.

Additionally, we will test our network on real lenses discovered by SuGOHI. We will also describe the pre-processing steps applied to these SuGOHI lenses to make them compatible with the inputs on which the network was trained. Our aim is to ensure that both the simulated and observational datasets reflect the complexities and noise characteristics expected in actual survey data, thereby enabling robust and generalizable model performance.

As we discovered in G25, the presence of lens light significantly influences the prediction of lens mass parameters, often biasing them toward values associated with the lens light profile. Therefore, accurately removing lens light becomes a critical pre-processing step. To address this, we process the SuGOHI lenses using the YattaLens (YL , Sonnenfeld et al. 2018) pipeline. This pipeline models and subtracts the lens light using an MCMC-based approach. In addition, YL generates an image segmentation map that identifies and distinguishes between lensed arcs and foreground objects. We use this map to mask out the foreground objects, retaining only the lensed arcs before feeding the images to the network.

2.1 Simulated datasets

Gravitational lensing represents a challenging inverse problem, where the objective is to reconstruct the properties of the lens mass distribution and the source from noisy observations of distorted background sources. The diversity in image configurations arises due to variations in the source's position relative to the lens and the complexity of the lensing potential. As a result, training convolutional neural networks (CNNs) for lens modelling needs access to a large and varied dataset of simulated lensing scenarios. Given that the

number of confirmed galaxy-scale lenses identified in the Hyper Suprime-Cam (HSC) Survey is only on the order of $O(100)$, which is not sufficient to sample the multi-dimensional parameter space, we produce a simulated training dataset of mock lenses to carry out the training process.

To generate these simulated systems, we utilize the SIMCT framework (More et al. 2016)[1], which is designed to create realistic gravitational lens images tailored to specific survey characteristics. Although originally developed for the Canada-France-Hawaii Telescope Legacy Survey (CFHTLS), we adjust its configuration parameters to match the observational properties of the HSC Survey, including depth, pixel scale, and point spread function (PSF). For a full description of the SIMCT methodology, we refer readers to More et al. (2016); here, we provide a brief overview of the relevant steps used to construct our training sample.

The pipeline adopts a hybrid image generation strategy, wherein simulated lensed arcs are superimposed on actual cutouts of candidate lensing galaxies, along with line-of-sight objects extracted from survey data. Background sources, such as galaxies or quasars, are modelled using analytic light profiles, with parameters sampled from realistic distributions reflecting empirical luminosity functions, redshift distributions, sizes, and spectral colors. Lens galaxy properties such as apparent magnitude, ellipticity, redshift, and orientation are converted into mass profile parameters under the assumption that light is correlated with mass, using scaling relations from the literature. The lens itself is modelled as a Singular Isothermal Ellipsoid (SIE), and an external shear component is introduced, sampled uniformly over a plausible range. Only those systems that meet predefined detectability thresholds (image separations exceeding 0.5 arcseconds and a second brightest image above a designated r-band magnitude limit) are retained in the final data set.

For complete details regarding the simulation parameters, input distributions, and adopted scaling relations, we refer the reader to Section 3 of Jaelani et al. (2023). The distributions of the lens mass parameters used in this study can be found in fig.2 of G25. To incorporate observational complexities, all simulated arcs are convolved with the co-added PSF profiles obtained from the HSC database at the location of each central lens galaxy, and Poisson noise is added in the g-, r-, and i-bands.

For training purposes, we however differ from the normal methodology and embed the noise-added, PSF-convolved lensed features on top of cutouts selected from HSC imaging data with size $17'' \times 17''$. The light from the lens galaxy can blend into the images of source galaxies making the inference of lens mass parameters difficult. The issue of lens light contamination and its impact on modelling accuracy has been discussed in Hezaveh et al. (2017), where the authors demonstrated that subtracting the central lens galaxy from high-resolution Hubble Space Telescope images significantly improved parameter inference. This concern is even more pronounced in ground-based data, where lower resolution exacerbates blending effects.

In this paper, we avoid this complexity and assume that lens light subtraction algorithms will be able to tackle this problem separately. Therefore, the cutouts which we choose to embed the lensed features contain no galaxy brighter than $i = 24.5$ magnitude within a central circular region of $5''$ radius. The resulting image set, referred to as the HSCempty sample, serves as a proxy for real lenses where lens light subtraction has already been performed and is perfect.

[1] https://github.com/anumore/simct

2.2 Real datasets

In addition to the simulated test set, we will evaluate our model on a sample of real lenses. This dataset comprises 182 strong lensing systems, classified as grade A and B, sourced from the SuGOHI survey. Given that we will train our network on the HSCempty sample, we need to make the input from these real lenses compatible with the network requirement by subtracting the lens galaxy.

We employ the YL pipeline for this task. The pipeline models the light profile of the foreground galaxy, subtracts it from the image, identifies potentially lensed features and fits a Singular Isothermal Ellipsoid (SIE) model to infer the lens model parameters. After subtraction, we further refine the images by removing residual foreground objects identified via segmentation maps produced by YL . The resulting lens-light-subtracted and cleaned images constitute our real-image test sample. For evaluation, we compare the predictions from our trained network with the corresponding parameters obtained from the YL pipeline.

3 NETWORK ARCHITECTURE AND TRAINING

The base of our BNN is the CNN that was developed in G25. We briefly summarize this CNN and the modifications we carried out in order to convert it to a BNN. The network adopted in G25 consists of seven convolutional blocks, each of which is alternately followed by average pooling layers. The network ends with two fully connected layers and an output layer. Each convolutional block includes a convolutional layer, followed by batch normalization, a Parametric Rectified Linear Unit (PReLU) activation, and a dropout layer with a 20 per cent dropout rate.

The first difference arises in the output layer. In G25, our output layer consisted of only the model parameters, while in this work we extend it to also include the covariance of these parameters. The second difference is related to the loss function we adopt. In contrast to G25, where the simple Mean-squared-error loss was used, we train the BNN using an alternative Negative Log-Likelihood (NLL) loss function. We optimize this objective function using the Stochastic Gradient Descent (SGD) techinique with a learning rate of 0.001.

The network is trained on multi-wavelength optical imaging, with input comprising the g, r, and i band images (101×101 pixels) of each strong lens system. We reserve 10 per cent of the training data for validation and train the model using a batch size of 64. To prevent overfitting, we adopt early stopping, terminating the training if the validation loss does not improve for 25 consecutive epochs. In practice, BNNs are more challenging to train than CNNs, often requiring a larger number of epochs to achieve convergence due to their increased complexity and stochastic nature.

We use the HSCempty sample for constructing the training set consisting of 60000 lenses out of which 10 percent are reserved for validation. The evaluation of the performance of the BNN is carried out on a reserved test sample of 20000 lenses.

The NLL loss function we adopt corresponds to a multivariate Gaussian distribution,

$$-\log \mathcal{L}_{\rm NLL} = \frac{1}{2}\log\left((2\pi)^n \det(\mathbf{\Sigma})\right) + \frac{1}{2}(\mathbf{y}-\boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{y}-\boldsymbol{\mu}) \quad (1)$$

where, n is the number of the lens and source model parameters, y are the true values of these parameters for the training set, and μ are the predicted values. The symbol Σ denotes the covariance matrix and consists of $n(n+1)/2$ additional numbers that the network is expected to learn about each system.

Mean squared error losses typically neglect the first term in the

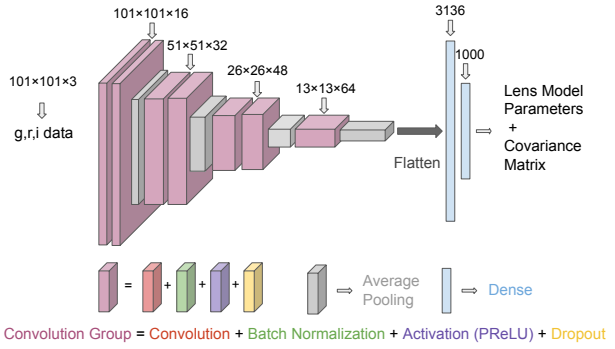


Figure 1. Network Architecture : The BNN architecture adopted in this work is composed of multiple convolutional blocks, each followed by average pooling layers to progressively reduce spatial dimensions. These are followed by two fully connected (dense) layers and an output layer that predicts the desired model parameters together with the covariance matrix.

above loss function. If we would have adopted a similar loss function while training the network to output the covariance matrix, then the network would simply adopt large values for the diagonal elements of the covariance matrix, driving down the loss function, even when the predictions do not match the truth. The first term in the NLL loss function penalizes this behaviour and thus brings a balance between matching the predictions, while keeping the determinant of the covariance small.

We will consider two different cases in our analysis. In Case A, we will consider key lens and source parameters which are the Einstein radius ($\theta_{\rm E}$), mass ellipticity components (e_x and e_x), source position (x and y) and its effective radius ($R_{\rm eff}$). For this case, we consider only a diagonal covariance matrix and the network outputs the parameters and the errors on these parameters. This setup requires around 800 epochs to train.

For Case B, we extend the network's predictive capability to output the entire covariance matrix. To avoid a large increase in the complexity, we restrict ourselves to only the lens mass model parameters in this case, namely, the Einstein radius $\theta_{\rm E}$ and the mass ellipticity components e_x and e_y as these parameters are expected to exhibit mutual correlations. The network in this case is trained for approximately 500 epochs.

We further simplify by asking the network to predict the diagonal elements of the covariance matrix, and the independent coefficients of the cross-correlation matrix $\mathcal{R}$, where

$$\mathcal{R}_{ij} = \frac{\Sigma_{ij}}{[\Sigma_{ii}\Sigma_{jj}]^{1/2}} \quad \forall i \neq j \tag{2}$$

To avoid numerical instabilities a small jitter term of 10^{-6} is added to the diagonals of the reconstructed covariance matrix before computing the NLL loss function.

4 RESULTS AND DISCUSSION

4.1 Case A: Lens and source model parameters and their errors

We first discuss our results for Case A, where we train our Bayesian Neural Network (BNN) using simulated HSCempty sample (60000 lenses) for which the ground truth is known for the lens mass model parameters Einstein radius, the two components of ellipticity, the source position, and the effective size of the source (defined as the half-light radius). The BNN is trained to predict these parameters alongwith the diagonal elements of the covariance matrix for each of these parameters, based on the loss function given by Eq. 1. The model is thus expected to learn to identify lensed features, and obtain a mapping from the lensed features in multi-wavelength images and infer physical parameters. The performance of the trained BNN on a simulated test sample consisting of 20000 lensed images is presented in Fig. 2.

In the upper row of Fig. 2 we present the network predictions for the parameters of the mass model for the lens system $\theta_{\rm E}, e_{\rm x}, e_{\rm y}$ plotted against their true values. The lower rows similarly show the network predictions for the source parameters, the positional offsets of the source with respect to the lens, and the effective radius of the source galaxy. In each of the panels the square root of the diagonal of the associated covariance matrix element is also plotted as an errorbar in the vertical direction, i.e., on the predictions.

We see that in the small $\theta_{\rm E}$ regime ($\theta_{\rm E} < 1.5''$), the network predictions agree fairly well with the true values, showing very little evidence for any bias and small uncertainties. At larger values of $\theta_{\rm E}$, the predictions exhibit increasing discrepancies, with the network systematically under-predicting the values and resulting in larger associated uncertainties. This is likely a consequence of having a smaller number of large $\theta_{\rm E}$ systems we have in our training sample. Since our simulated dataset is designed to reflect the real distribution of lenses in the Universe, systems with smaller $\theta_{\rm E}$ are far more prevalent and the network demonstrates reliable performance in this parameter range and successfully recovers the Einstein radius for the majority of lensing systems.

In G25, we observed similar behaviour and applying sample weights led to a slight improvement in network performance for systems with larger $\theta_{\rm E}$. In G25, we used Keras's built-in MSE loss function, which allowed us to pass sample weights directly via an argument. However, our current custom loss function does not yet support sample or class weights. Extending our custom loss to support sample and class weights, similar to built-in Keras losses is an important direction for future work.

For the mass ellipticity components e_x and e_y, we observe a positive correlation between the network predictions and the truth, albeit with an under prediction of the overall amplitude when the ellipticity components deviate from zero. Given the realistic distribution of ellipticities in galaxies expected in the Universe, there is a large number of systems in the training sample with either of the components being close to zero. Thus a large fraction of the predictions seem to match the expectation.

The network is also able to recover the positional offsets in the x and y directions in a fair manner, as seen from the first and second panels of the lower row of Fig. 2. The location of the source affects the multiplicity of the images that are formed as well as the anisotropic magnification that the images experience and is an indication that the lensed features are being learnt appropriately. The effective radius of the source $R_{\rm eff}$ displayed in the third panel again shows a positive correlation with the truth, although the network begins to under-predict this parameter at higher values where the discrepancies become more pronounced. Once again, this trend is likely a result of the lower representation of sources with large $R_{\rm eff}$ in the training set, which could possibly be slightly improved with sample weights.

The errorbars on each of these parameters, span the overall range of the predictions given the true values, thus characterizing the accuracy of the predictions in a fair manner. The loss function we adopt naturally avoids errors that are over or under-inflated. Although we

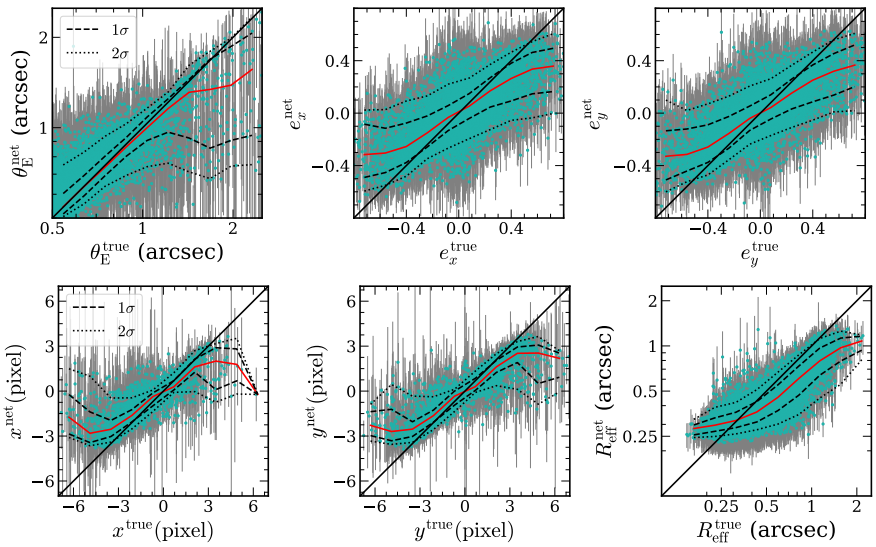


Figure 2. Case A: The performance of our network, trained on the HSCempty dataset comprising 60,000 simulated lens images, is evaluated on a separate test set of 20,000 lenses. The upper row shows the network's predictions for the mean and associated uncertainties of the lens mass parameters: the Einstein radius ($\theta_{\rm E}$) and the ellipticity components (e_x and e_y), compared against their true input values. The lower row displays analogous results for the source parameters, including the source position coordinates (x, y) and the effective radius ($R_{\rm eff}$). The red curve represents the median, while the black dashed and dotted curves correspond to the bounds of the 68 (1σ) and 95 (2σ) percentiles, respectively.

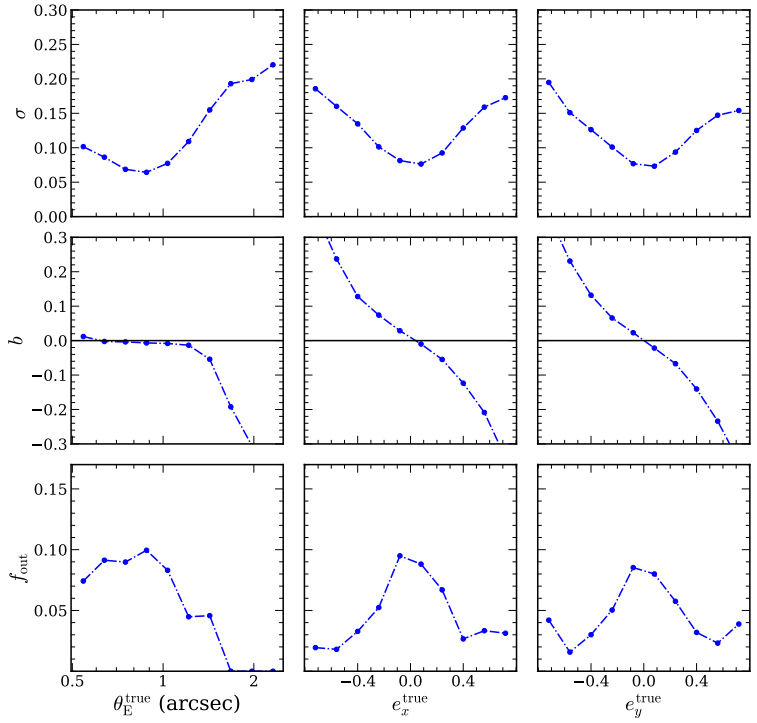


Figure 3. Performance of our network on the HSCempty test sample (Case A). The columns correspond to the lens parameters $\theta_{\rm E}$, e_x, and e_y, respectively, while the rows represent the evaluation metrics: standard deviation (σ), bias (b), and outlier fraction ($f_{\rm out}$), as defined in the Section 4.

acknowledge that the accuracy of the errors has not been directly tested in this current work.

We present a statistical evaluation of the network's performance by computing the bias, scatter and the outlier fraction for each of the lens and source parameters. The range of true values for each parameter is divided into 10 bins, logarithmically spaced for the Einstein radius ($\theta_{\rm E}$) and effective radius ($R_{\rm eff}$), while linearly spaced for the ellipticity components (e_x and e_y) and the source position (x and y). For $\theta_{\rm E}$ and $R_{\rm eff}$, we compute the relative difference, $\theta_{\rm E}^{\rm net}/\theta_{\rm E}^{\rm true} - 1$, $R_{\rm eff}^{\rm net}/R_{\rm eff}^{\rm true} - 1$ and use its standard deviation and median to estimate

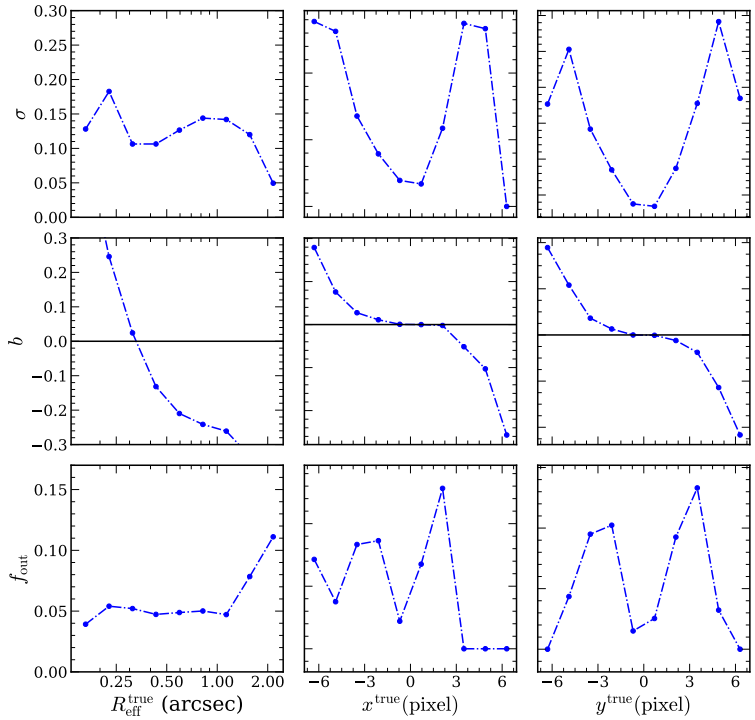


Figure 4. Performance of our network on the HSCempty test sample (Case A). The columns correspond to the source parameters $R_{\rm eff}$, x, and y, respectively, while the rows represent the evaluation metrics: standard deviation (σ), bias (b), and outlier fraction ($f_{\rm out}$), as defined in the Section 4.

the scatter (σ) and bias (b), respectively. For e_x, e_y, x and y, scatter and bias are calculated similarly, using the absolute differences $e_x^{\rm net} - e_x^{\rm true}$, $e_y^{\rm net} - e_y^{\rm true}$, $x^{\rm net} - x^{\rm true}$ and $y^{\rm net} - y^{\rm true}$. Finally the outlier fraction ($f_{\rm out}$) is defined as the fraction of points in each bin that fall outside a region spanning twice the 1σ range around the median. If the error distribution is Gaussian around the median, we expect approximately 5% of the data points to lie outside this range. Any deviation of the outlier fraction, e.g. above (below) this 5 percent expectation, corresponds to larger (smaller) tails of the distribution.

The performance of our network for Case A, using the statistical quantities defined above (standard deviation (σ), bias (b) and outlier fraction ($f_{\rm out}$)) is shown in Fig. 3 and Fig. 4 for the lens and the source parameters, respectively. The three columns of Fig. 3 correspond to the lens parameters $\theta_{\rm E}$, e_x, and e_y, respectively. Similarly, the three columns of Fig. 4 correspond to the source parameters $R_{\rm eff}$, x, and y, respectively.

The bias in the predicted parameter values for some of the parameters may seem puzzling at first glance. Given that our eye can fit a line and correct for this bias, one may be tempted to wonder why the network is unable to correct for this bias. We have checked that even in the case of the training sample, we observe the same sort of bias in these predictions. Given this we even considered a post-facto bias correction which can be estimated from the training sample data which could potentially be applied to the predicted parameter values. In order to correct the bias, we need to estimate the bias as a function of the network predicted values. After all, in real data neither the network nor we will have access to the true values.

Our investigations, however, showed that the bias corrections when estimated as a function of predicted parameter values appear quite low. In order to understand this, one needs to transpose each of the panels shown in Fig. 2. Once transposed, the distribution of the true parameter values as a function of the predicted parameter values is affected by the inherent distribution of the true parameters (in some sense the prior distribution of the parameters). Let us consider for example the case of the Einstein radius $\theta_{\rm E}$. At large values of network predicted $\theta_{\rm E}$, we see that there is a large number of points which lie on the one is to one line (at lower true $\theta_{\rm E}$), and thus the median is entirely driven by these points. If we were to correct for a large amount of bias as a function of the network predicted parameter, we would end up with incorrect predictions for a large number of these systems. Similarly, for the ellipticity components of the mass mdoel, the large set of points near $e_{\rm x}$ and $e_{\rm y}$ equal to zero, would be affected by any bias corrections made as a function of the predicted parameters.

4.2 Case B: Lens model parameters and their covariance

For Case B, we trained our network architecture using the same HSCempty simulated lensing dataset, but in contrast to Case A, our output layer focuses only on the key lens mass parameters: the Einstein radius and the two components of ellipticity, and the covariance matrix of these parameters. Given that our aim was to also estimate the covariance matrix, exclusion of the source parameters could potentially reduce the complexity of the network. The prediction of the mean parameters together with their covariances is a significant difference compared to previous cases of parameter estimation for gravitational lenses considered in the literature, as this would allow the network to model the underlying joint posterior distribution of the parameters. This would naturally account for parameters correlations that arise in lens modelling and would significantly reduce the computational challenges associated with Monte-Carlo sampling.

The three panels in the upper row of Fig. 5 show the network predictions for the mean and uncertainty of the mass model parameters $\theta_{\rm E}$, e_x, and e_y compared to their true values. These results are quali-

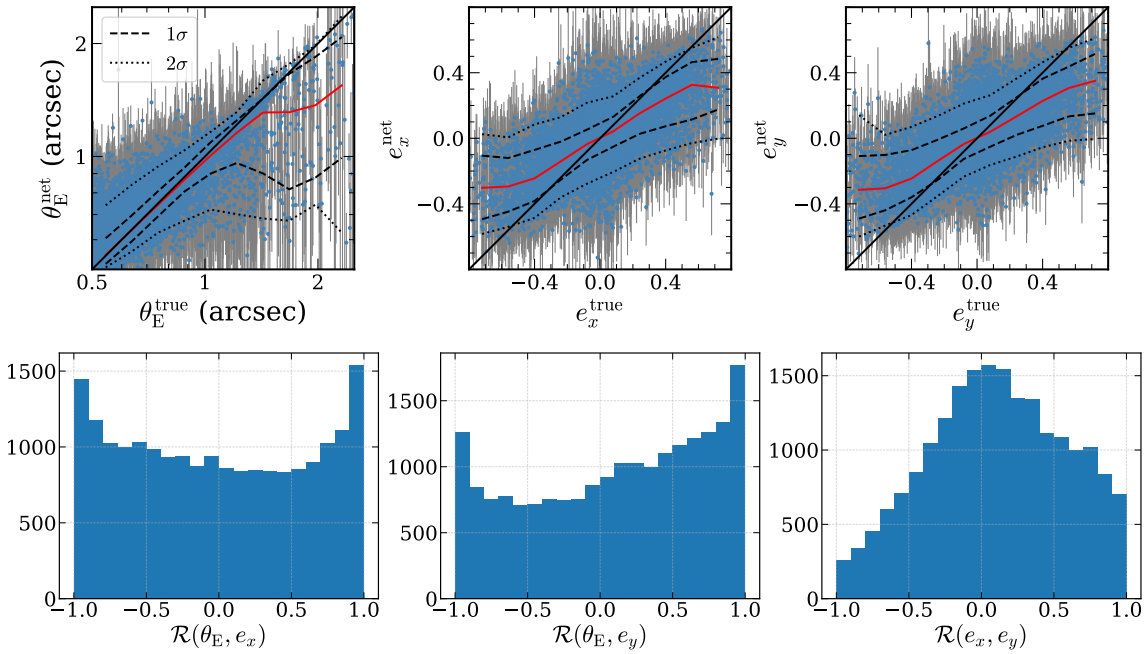


Figure 5. Case B with simulated test sample: We evaluate the performance of our network, trained on the HSCempty dataset comprising 60,000 simulated lensing images, using an independent test set of 20,000 simulated lenses. The upper row shows the predicted mean values and corresponding uncertainties for the lens mass parameters, Einstein radius ($\theta_{\rm E}$) and ellipticity components (e_x and e_y) in comparison to their true input values. The red curve represents the median, while the black dashed and dotted curves correspond to the bounds of the 68 (1σ) and 95 (2σ) percentiles, respectively. The lower row displays normalized correlation coefficient among these parameters as predicted by the network.

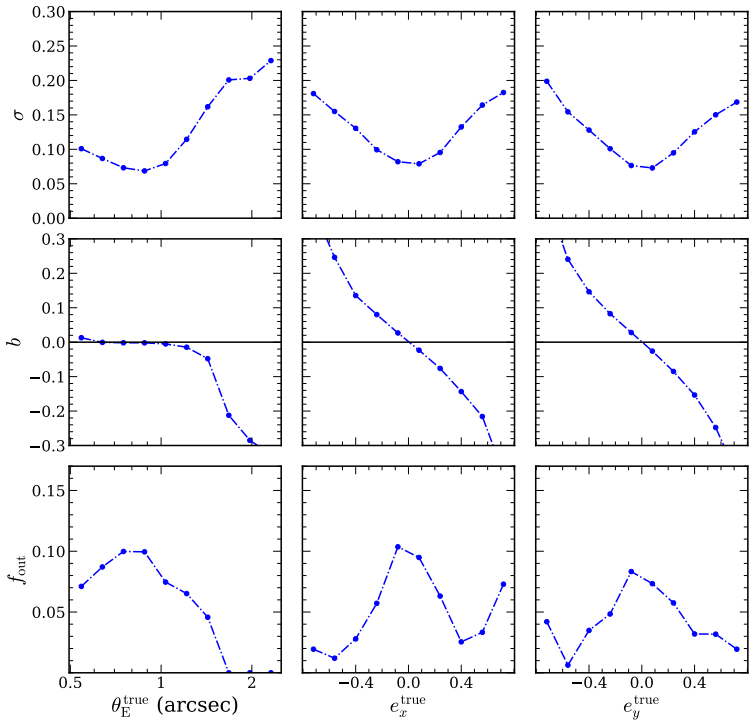


Figure 6. Performance of our network on the HSCempty test sample (Case B). The columns correspond to the parameters $\theta_{\rm E}$, e_x, and e_y, respectively, while the rows represent the evaluation metrics: standard deviation (σ), bias (b), and outlier fraction ($f_{\rm out}$), as defined in the Section 4.

tatively similar to those seen for Case A (upper row of Fig. 2) and our comments from the previous section also hold in this case. The lower panels show the cross-correlation coefficients R_{ij} obtained from the network for each of the parameter combinations. Interestingly we find fairly uniform correlation coefficients between the Einstein radius and the ellipticity components with a modest increase in the cross-correlation coefficients closer to ±1. Although we have not explored it here, it would be interesting to explore systems which show such large cross-correlation to figure out if they correspond to systems with certain features and indeed represent true degeneracies in the parameters.

The performance of our network for Case B, evaluated using the statistical quantities, standard deviation σ, bias b, and outlier fraction $f_{\rm out}$ are shown in Fig. 6 with the three columns corresponding to the lens parameters $\theta_{\rm E}$, e_x, and e_y, respectively. Once again we see a similar pattern compared to Case A, with biases seen to increase where the training sample is sparsely available, i.e., for large $\theta_{\rm E}$ and for large absolute values of ellipticity components.

4.3 Comparison on real lens systems

Finally, we also applied the trained model from Case B to the sample of 182 real SuGOHI lenses. These samples have been modelled previously using the MCMC-based modeling pipeline, YL in G25. Given that our model was trained on the HSCempty sample, we used the lens light subtraction algorithm from YL in order to obtain the lensed images of the source devoid from the lens light. This allows us to compare the network's predictions with those obtained from the MCMC modelling as well as the associated covariance matrices.

We show the results of this comparison in Fig. 7. The top right panel shows that we find an excellent agreement between the inferred values of the Einstein radius based on either of the two methods, a result which is quite encouraging. The errors on both axes represent the corresponding errors by each of the pipelines. A closer look reveals a small but persistent trend of certain lens systems to show slightly larger values of the Einstein radius in the MCMC modelling compared to the network prediction. Such systems would be worth exploring for signs of imperfect modelling of the lens light (see G25).

The success in the predictions of the Einstein radius of real lens system does not seem to carry over as well to the predictions for the ellipticity components e_x and e_y. Although there is a positive correlation between the predictions of the ellipticity components between the two methods (see a large number of points clustered around the one is to one line), there are quite a few systems where higher ellipticity systems as flagged by YL are given values of ellipticity components close to zero giving an apparent flat behaviour in the comparison of the ellipticity components.

The lower panels show the cross-correlation coefficients predicted by the network. We do not entirely see the trends we had previously seen of almost uniform cross-correlations between $\theta_{\rm E}$ and the ellipticity components. It is likely that the distributions of the observed lenses is different from the underlying true distribution of lenses, given the selection biases introduced by lens finding.

We compare the uncertainties from our network with those derived from YL and these are presented in Fig. 8. The errors in $\theta_{\rm E}$ appear largely consistent between the two methods. The errors on e_x, e_y also show some correlation but quite a large amount of scatter. An exploration of signs of imperfect modelling of lens light is warranted given our results to understand if it results in the increased scatter seen in these figures.

Finally, we also compare the cross-correlation coefficients from the posterior distributions obtained from the MCMC chains from YL for a subset of the real systems in Fig. 9. The posterior samples from YL given the lensed images are shown as points in each panel of figure. The corresponding 68 per cent confidence contours are also overlaid on top in cyan color. The equivalent contours generated from the network's predicted covariances are shown in red. Although some systems appear to show a reasonable agreement between the two methods, there are quite a few systems where the covariance between the two methods do not agree.

4.4 Discussion and directions for future

There are differences observed in parameter inferences as well as covariances between traditional MCMC modelling methods compared to machine learning methods documented here and G25 as well as in (Schuldt et al. 2023b). Given these differences, an understanding of the features that the network latches on to for systems that show significant differences need to be explored in detail. The extracted features at each layer of the CNN which is at the heart of our BNN needs to be explored for sensitivity to the parameters.

For cases where the parameter inferences agree but the covariances are in disagreement, one could investigate the predicted models along the different degeneracy directions by the two methods. This could help understand the differences in features to which the network is sensitive to compared to the features that the lens modelling techniques devote their attention.

Yet another technique to explore the accuracy of the errors is to simulate a large number of lens systems and investigate probability-probability plots corresponding to each of the parameters in order to understand the reliability of the errors. Explorations of such kind would require a large number of traditional modelling runs to be performed on simulated lens systems. Our network could however be relatively quickly tested on such large number of simulated systems to assess the reliability of its errors. Such a test is useful to understand the diagonals of the covariance matrix, and we plan on investigating generalizing such methods to multi-dimensional parameter space in future.

Although the BNNs used in this work have similar architecture as that of the CNN we explored in G25, we trained the new network entirely from scratch. In future, a transfer learning approach may help to speed up some of the explorations further and perhaps focus on training to infer the covariances with the final dense layers of the network.

5 SUMMARY

In this study, we developed, trained and evaluated the performance of a Bayesian Neural Network to predict the posterior distribution of lens model parameters given multi-wavelength imaging data of strong lens systems discovered in deep ground-based imaging surveys such as the HSC. Our Bayesian neural network was built on top of a CNN and was supplied with a custom loss function designed to estimate both the parameters as well as the covariances of the inferred parameters.

Our results can be summarized as follows:

- We trained a BNN to infer the parameters and associated uncertainties in model parameters from multi-wavelength images. We quantitatively evaluated the performance of the network to infer the parameters Einstein radius and the ellipticity of the mass distribution in addition to the positional offsets and the effective radius of the source and found that the network was able to learn these parameters.

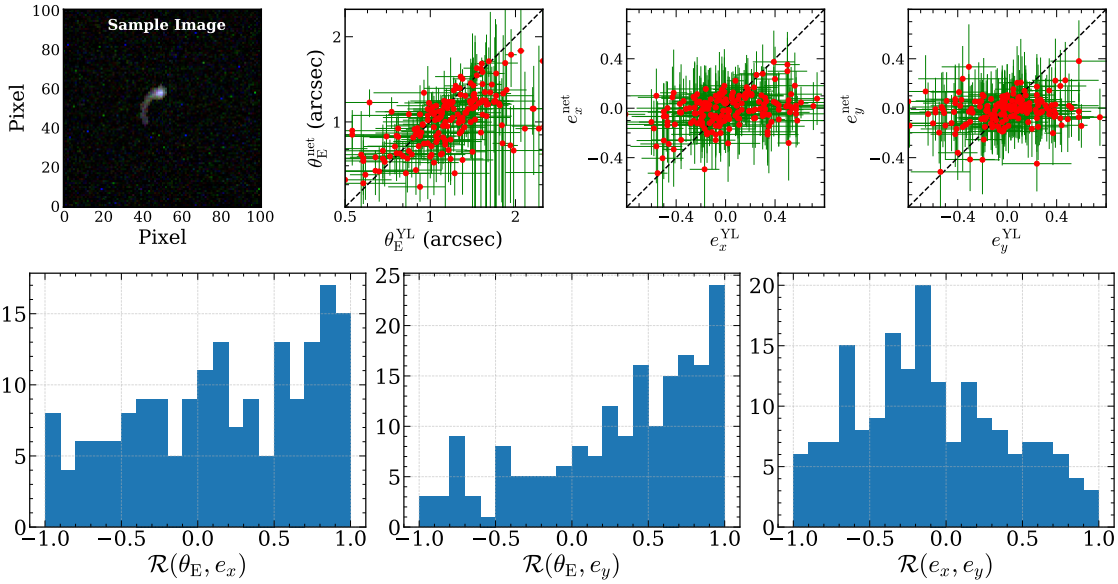


Figure 7. Case B with SuGOHI test sample: We assess the performance of our network, trained on the HSCempty dataset containing 60,000 simulated lensing images, by applying it to a test sample of 182 real lenses from SuGOHI. The top row presents the network's predicted mean values and associated uncertainties for the lens mass parameters, Einstein radius ($\theta_{\rm E}$) and ellipticity components (e_x and e_y) and compares them with the estimates obtained from the YL modelling framework. The bottom row illustrates normalized correlation coefficients among these parameters as predicted by the network.

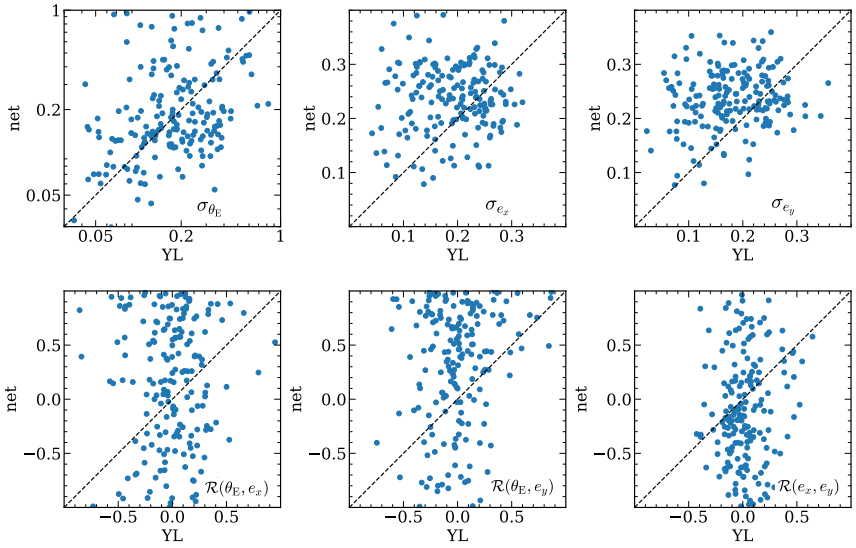


Figure 8. The top row compares the uncertainties corresponding to the lens mass parameters, the Einstein radius ($\theta_{\rm E}$) and the ellipticity components (e_x and e_y) as predicted by our network for Case B on the SuGOHI test sample with those obtained using YL . The bottom row compares the normalized correlation coefficients among these parameters as predicted by the network with the estimates from YL .

- We also designed and developed and evaluated the performance of an upgraded BNN which infers a subset of these parameters along with the covariance of uncertainties designed to mimic the output of traditional MCMC modelling methods that obtain a posterior distribution of parameter samples given the data.
- In a large range of parameter space, we saw positive correlations between the predicted parameters and their true values. However the output of the covariance matrix of the inferred parameters needs to be further stress tested for accuracy.
- We further applied the BNN to a set of real galaxy-galaxy lenses selected from the HSC survey, and compared the performance of our network with YL , a traditional lens modelling method.
- The Einstein radius remains the most robust parameter that the two methods agree on. Ellipticity components of the mass model show variation between the two methods, another area ripe for further exploration.

Overall, our results show that BNNs are capable of efficiently inferring both point estimates and uncertainties of lens model parameters from imaging data by capturing the full joint posterior distribution. These networks combine the fast computation and scalability inher-

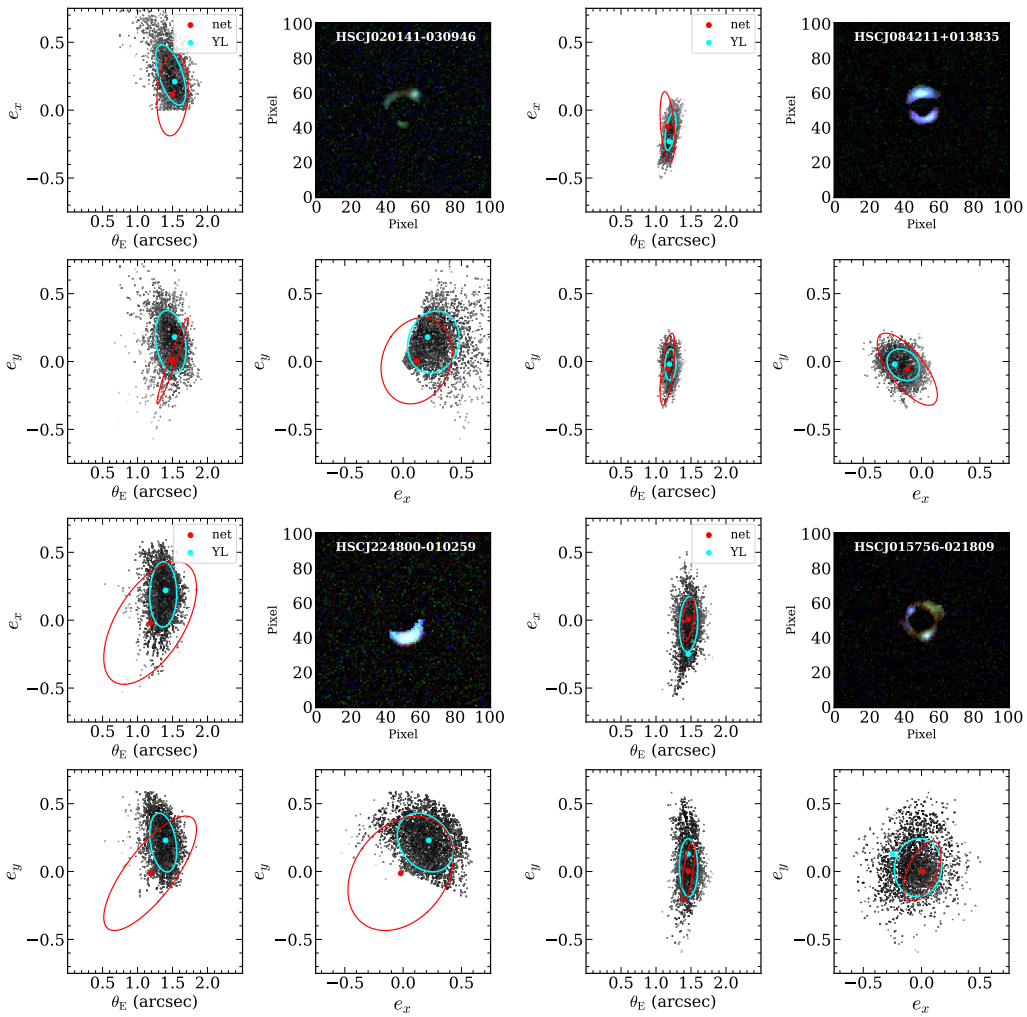


Figure 9. A visual comparison of the covariance matrices derived from our network and from YL is presented. The black scatter points represent the MCMC chains from YL , with the cyan ellipse indicating the corresponding 68 per cent confidence contour. The red ellipse denotes the 68 percent confidence region as predicted by the covariance matrix output of our network. We illustrate these comparisons for four representative lenses from the SuGOHI sample.

ent to machine learning with the robust uncertainty quantification traditionally provided by MCMC methods. This advancement represents an important contribution towards efficient and probabilistic lens modelling, making BNNs particularly well suited to handle the large influx of lenses anticipated from upcoming surveys such as LSST.

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DATA AVAILABILITY

The SuGOHI [2] lenses and the YL [3] pipeline used in this paper are publicly available. The CNN code and simulated images can be made available upon a reasonable request to the corresponding author.

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